

User Study

Instructions for Grading. **PLEASE READ CAREFULLY.**

Overview

Imagine you are a user interacting with a chatbot service (e.g., Gemini, ChatGPT, or DeepSeek) for help with programming tasks. On each page, you will see:

- **A Question:** This could be a vague user query or a code snippet requiring clarification or completion. Questions containing code will be displayed in code blocks for clarity.
- **Two Responses:** labeled Answer A and Answer B

Your Task

Please evaluate Answer A and Answer B based on the following criteria:

- **Precision and Focus:** How directly the response targets the question's core issue, using clear guidance, examples, or code.
- **Contextual Fit:** How well the response matches the user's programming goal (e.g., debugging, feature implementation).
- **Answer Faithfulness:** How effectively the response provides actionable steps or code to achieve the user's desired outcome.
- **Correctness:** Correctness in LLM responses refers to the accuracy and factual validity of the information, code, or guidance provided, ensuring it is free from errors and aligns with the user's query and intended outcome.

How to Rate

For each question, you will provide two types of ratings:

1. Preference Rating (1–5 scale):

- **"1"**: Answer A is better, based on the criteria above.
- **"5"**: Answer B is better, based on the criteria above.
- **"3"**: Both answers are equally good, based on the criteria above.

Note:

- **Incomplete Responses:** Some responses may appear cut off due to a 256-word limit imposed on the chatbots. Please grade only the content provided, ignoring any abrupt endings or non-English text (e.g., stray characters).
- **Code Formatting:** Questions and responses containing code will be presented in code blocks or artifact tags for clarity. Ensure you review these carefully for correctness and relevance.
- **Consistency:** Try to apply the same standards across all evaluations to ensure fair comparisons.

Question 1

Python

```
import pandas as pd
df = pd.read_csv('data.csv')
# TODO: Clean data
print(df.head())
```

Answer A:

“Extra context:

User Question:

Python

```
import pandas as pd

df = pd.read_csv('data.csv')

# TODO: Clean data
print(df.head())
```

AI clarification question: *What specific data cleaning or analysis are you hoping to perform on the DataFrame `df` after reading it? For example, are you looking to handle missing values, convert data types, or perform a different transformation?*

User follow up response: *I want to remove rows with missing values and convert a column to integers.”*

AI Final answer:

Okay, that's a good starting point. You want to remove rows with missing values and convert a specific column to integers. Here's the code to accomplish that:

Python

```
import pandas as pd

df = pd.read_csv('data.csv')

# Remove rows with missing values
df = df.dropna()

# Convert the 'column_name' (replace 'column_name' with the actual name of the
column you want to convert) to integers
try:
```

```
df['column_name'] = df['column_name'].astype(int)
except KeyError:
    print("Error: Column 'column_name' not found in the DataFrame.")
except ValueError:
    print("Error: Cannot convert 'column_name' to an integer. The column may contain non-numeric values.")

print(df.head())
```

Answer B:

```
Python
import pandas as pd

df = pd.read_csv('data.csv')

# TODO: Clean data
print(df.head())
```

Explanation:

1. **import pandas as pd**: Imports the pandas library, which is essential for working with data in Python.
2. **df = pd.read_csv('data.csv')**: Reads the data from a CSV file named 'data.csv' into a pandas DataFrame called **df**.
3. **print(df.head())**: Prints the first 5 rows of the DataFrame **df**. **df.head()** is a method that returns a view of the first 5 rows of the DataFrame. It's a quick way to inspect the data and see if it looks correct.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):1

Contextual Fit (1-5):1

Answer Faithfulness (1-5):1

Correctness (1-5):1

Question 2

```
HTML
<html>
<head>
<title>My Page</title>
</head>
<body>
<h1>Welcome to my page
<p>This is a paragraph.
```

Answer A:

```
HTML
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <h1>Welcome to my page</h1>
  <p>This is a paragraph.</p>
</body>
</html>
```

Explanation:

This is a basic HTML document.

- **<html>**: The root element of the HTML page.
- **<head>**: Contains meta-information about the HTML document.
- **<title>**: Specifies a title for the HTML page.
- **<body>**: Contains the visible page content.
- **<h1>**: Defines a level 1 heading.
- **<p>**: Defines a paragraph.

Answer B:

“Extra context:

User Question:

```
HTML
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <h1>Welcome to my page
  <p>This is a paragraph.
```

AI clarification question: What is the primary purpose or goal of this page?

User follow up response: I'm creating a blog page to display my posts.”

AI Final answer:

Okay, here's a breakdown of the page's purpose and potential improvements, based on your description:

Primary Purpose:

The primary purpose of this page is to serve as a **blog**. It's designed to display posts and welcome visitors.

Analysis:

- `<h1>Welcome to my page</h1>`: Main heading stating the page's purpose.
- `<p>This is a paragraph.</p>`: Introductory paragraph.

Potential Improvements:

- Add more specific content about the blog's topics.
- Consider a call to action, like inviting users to subscribe or read recent posts.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5):5

Answer Faithfulness (1-5):5

Correctness (1-5):5

Question 3

I'm not sure how to structure my project.

Answer A:

Okay, let's tackle this! It's completely normal to feel unsure about project structure. To help me give you the *best* advice, I need a little more information about your project. Let's start with some questions:

1. What kind of project is it? (Personal, work, academic, business, etc.)
2. What's the overall goal of the project?
3. What technologies or programming languages are you using?
4. Are you working alone or with a team?
5. What have you tried so far?

Answer B:

"Extra context:

User Question: *I'm not sure how to structure my project.*

AI clarification question: *To help me suggest a good structure for your project, could you tell me a little more about what kind of project you're working on? For example, what's the project about, and what technologies are you using?*

User follow up response: *I'm building a Node.js web app, and I don't know how to organize my files and folders."*

AI Final answer:

Great! Building a Node.js web app is a common project. Here's a recommended directory structure:

```
HTML
my-node-app/
├─ package.json      # Project metadata and dependencies
├─ src/              # Source code
│   └─ app.js         # Main application file
│   └─ routes/        # API route handlers
│   └─ models/        # Data models
│   └─ controllers/   # Request logic
│   └─ views/         # Templates (if using a view engine)
├─ public/           # Static files (CSS, JS, images)
└─ README.md         # Project documentation
```

This structure makes your project organized and scalable.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5):5

Answer Faithfulness (1-5):5

Correctness (1-5):5

Question 4

```
JavaScript
function fetchData(url) {
  // TODO: handle errors
  let response = await fetch(url);
  return response.json();
}
```

Answer A:

```
JavaScript
async function fetchData(url) {
  try {
    const response = await fetch(url);
    if (!response.ok) {
      throw new Error(`HTTP error! Status: ${response.status}`);
    }
    const data = await response.json();
    return data;
  } catch (error) {
    console.error("Error fetching data:", error);
    return null;
  }
}
```

Answer B:

“Extra context:

User Question:

```
JavaScript
function fetchData(url) {
  // TODO: handle errors
  let response = await fetch(url);
  return response.json();
}
```

AI clarification question: *What kind of data are you expecting to be returned from the URL? (e.g., JSON, HTML, XML, or something else?)*

User follow up response: *I'm expecting JSON data with a list of products from an e-commerce API."*

AI Final answer:

You are correct! Based on your description, you're expecting JSON data-specifically, a list of products. Make sure your error handling checks for response status, and that you parse the JSON correctly. If you want to process the product list further, you can iterate over the array and display product details in your app.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5):5

Answer Faithfulness (1-5):5

Correctness (1-5):5

Question 5

My program isn't working as expected.

Answer A:

Okay, I understand you're having trouble with your program. That's frustrating! To help me understand the problem and offer the best advice, I need more information:

1. What language is your program written in?
2. What is the program supposed to do?
3. Can you share a snippet of the code?
4. What's not working as expected?
5. Are there any error messages?

Answer B:

"Extra context:

User Question: *My program isn't working as expected.*

AI clarification question: *Could you tell me a little more about what's happening and what you expect your program to do? For example, what language is your program written in, and what specific issue are you seeing?*

User follow up response: *My C# program is supposed to display a form, but the form is blank."*

AI Final answer:

A blank form in a C# program is a common issue. Make sure you have added controls (like labels, buttons, etc.) to the form in your code, and that you're calling `InitializeComponent()` in your form's constructor. If you're using Windows Forms, check that your controls are added to the form's Controls collection, and that the form is being shown with `Application.Run(new Form1());`. If you share your code, I can help pinpoint the problem!

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5):5

Answer Faithfulness (1-5):5

Correctness (1-5):5